TAUREX DRILL BITS
POWERED BY: BITVISION™

# 7.875″ TSd516 S15809 EDR Analysis

SRH G 1448HA

Apache, Cactus 172

# Overview

- Cactus 172 used Taurex 7.875" TSd516 (S15809) on SRH G 1448HA to complete of the curve (407ft) and drill the lateral in a WCA target
  - S15809 drilled 5481 ft of lateral before being pulled early
  - At surface, the bit was rung out on the nose & shoulder
  - This report attempts to find the root cause of the early pull and dull condition

On-Bottom Parameters:

| | | ROP - Fast | Bit Weight | Top Drive RPM | Flow In Rate |
|---|---|---|---|---|---|
| HOLE SECTION | RIG STATE | Median | Median | Median | Median |
| LATERAL | ROTATE | 438.7 | 51.1 | 70.4 | 549 |
| | SLIDE | 45.4 | 50.9 | 6.4 | 549 |

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# Dull Condition: S15809

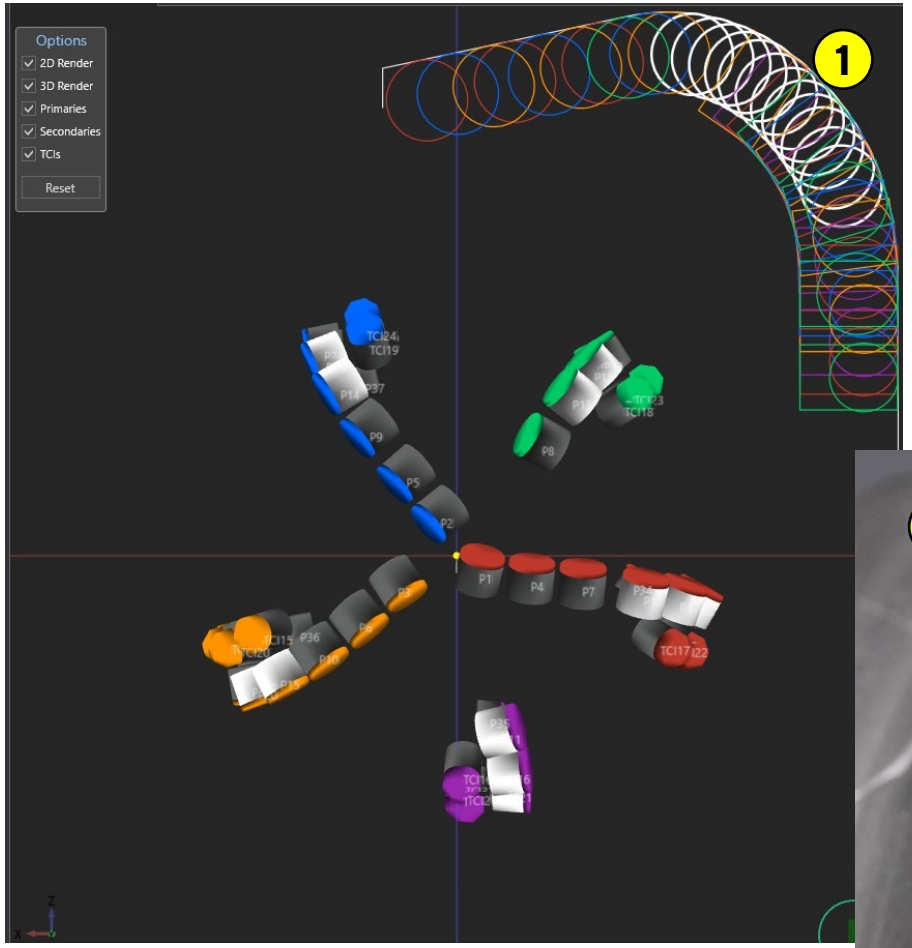
# Forensic Dull Analysis

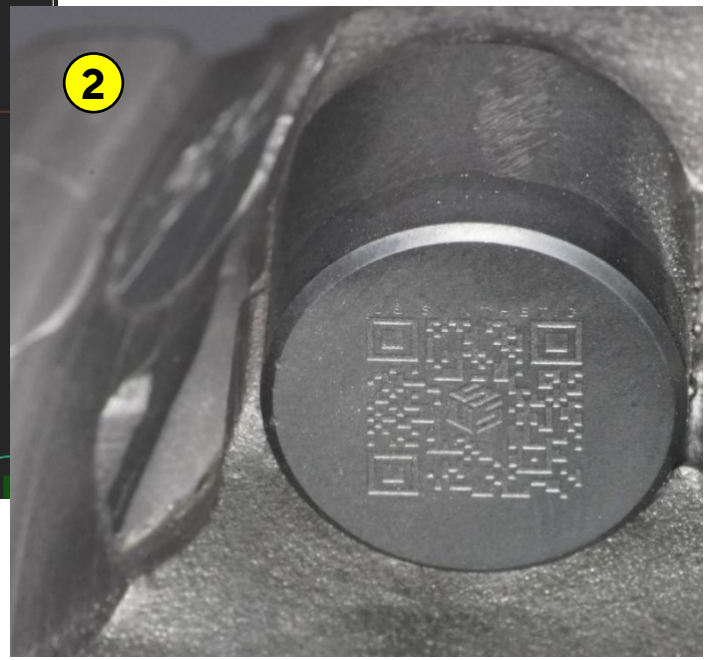
## Observations

1. Ringout occurs outside of the nose of the bit (damaged cutters highlighted in white)

- Outer nose & shoulder damage can manifest from interbedded drilling, particularly when exiting a harder formation and moving into a softer one

2. Little to no damage on cutters adjacent to the ringout, suggesting rapid failure progression → this is consistent with the EDR

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# EDR Overview: Lateral Run (By Depth) 8165’ to 13616’

Rotate

Slide

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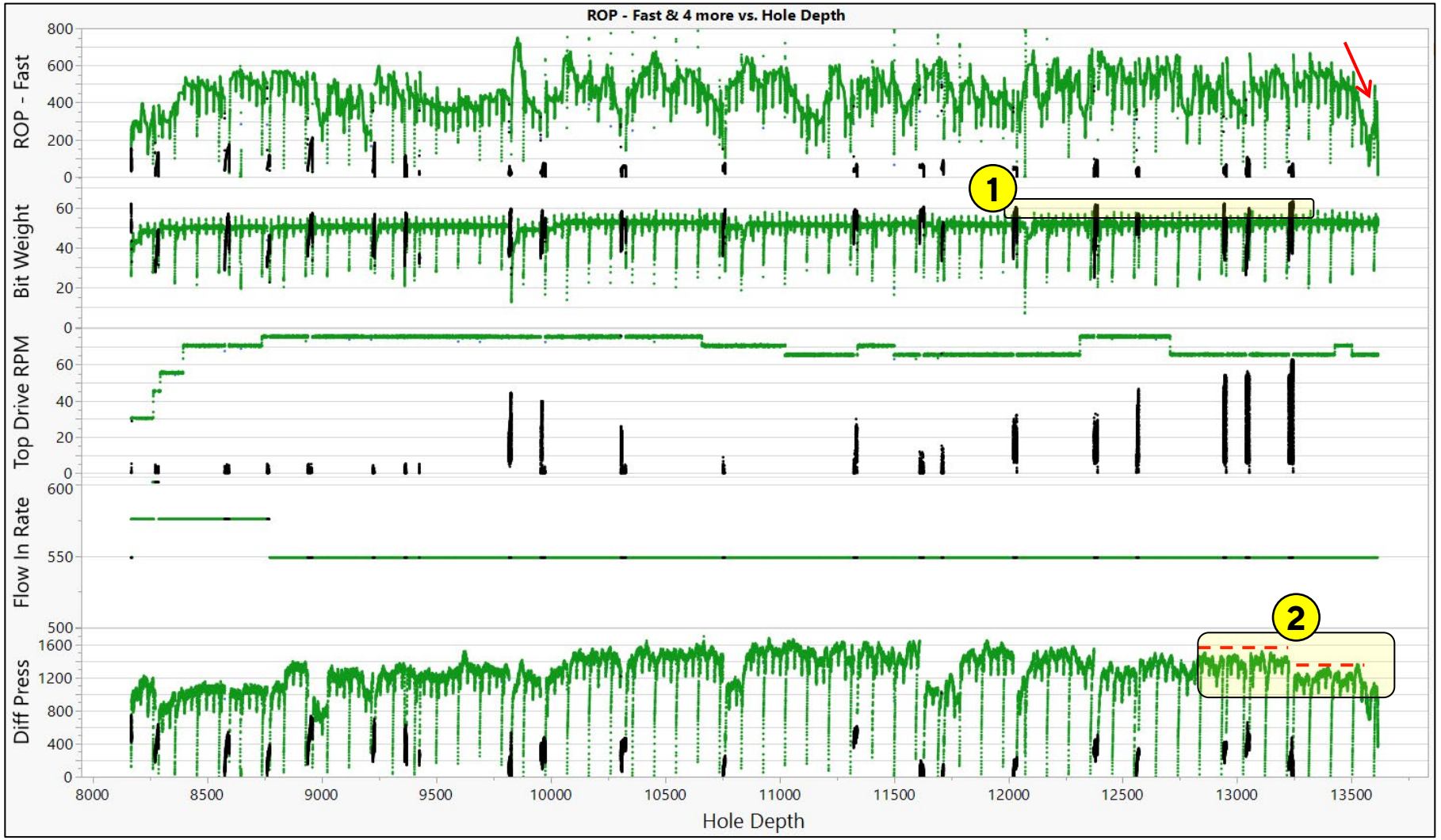
## Observations

1. During the last several slides, the WOB used to slide was ~60klbs while the drilling WOB was ~50klbs
2. Following the last slide, there is a drop in diff from ~1500 PSI to ~1200 PSI
3. During the final stand, ROP drops from 600 ft/hr to 100ft/hr then drops to 20ft/hr in the following stand

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# EDR: Last 3 Slides (By Time) ~12800ft to ~13400ft

**Rotate** | **Slide** | **Off Bottom**

Off-bottom values are determined by the "Bit Status" exported from Totco (Bit Depth < Hole Depth)

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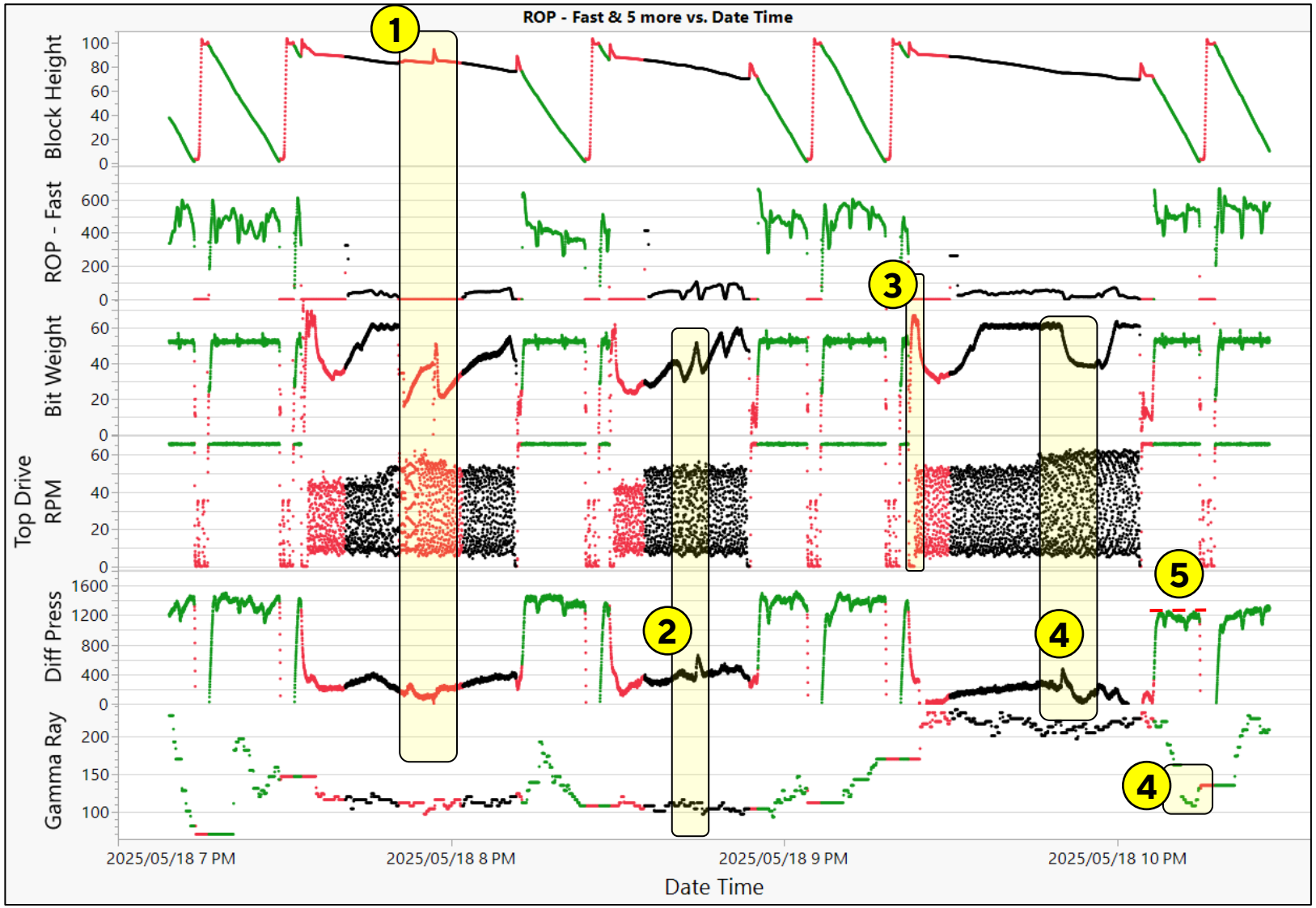
1. During the third to last Slide, ROP and diff drop with WOB at 60klbs, and the BHA is lifted off bottom while rocking pipe (~10 - 60RPM)
   - ROP and Diff recover in the remainder of the slide
2. During the second to last slide, diff spikes as WOB is increased to 50klbs, and WOB drops to ~35klbs
3. At the start of the final slide, while off bottom, a WOB of 70klbs is measured with the rotary at 0 RPM
4. Diff spikes during this last slide and WOB drops from 60klbs to 40klbs

   Gamma measurement drops ~50ft after this diff spike
   - Considering the gamma offset, the drop in gamma may be associated with this event (depth on next slide)
5. Following these slides, Diff falls to 1200 PSI, indicating the bit may have been damaged here

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# EDR Analysis : Last Slide/Gamma drop

Rotate

Slide

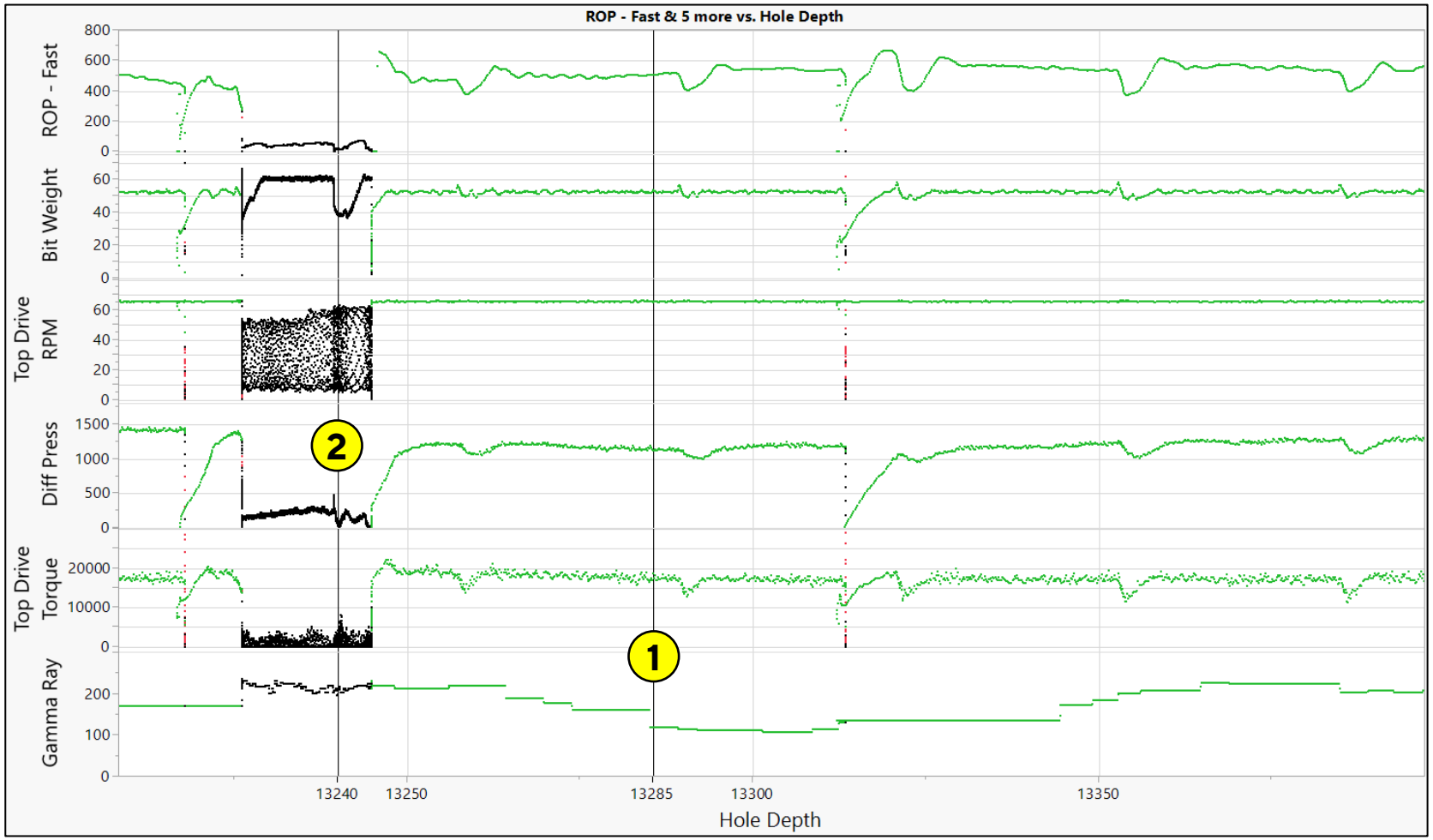
## Observations

1. There is a drop in gamma measured ~13285ft

   The MWD offset from the bit is ~50ft (provided in the BHA)

2. The drop in gamma coincides with the differential pressure spike ~ 13235ft

   Differential pressure drops after this slide despite similar parameters indicating the bit may have been damaged here

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# Conclusion

- Damage to S15809 was isolated to the nose of the bit
- EDR analysis showed ROPs of 600ft/hr until the final stands indicating damage occurred rapidly
- Interbedded formations and transitional zones (especially soft to hard) seen in the final slide can cause the damage seen at the apex of the TSd516 bit
- It is likely that the cutting structure was damaged during final slides, and rapidly progressed as the bit drilled another ~400ft before ROP dropped off completely

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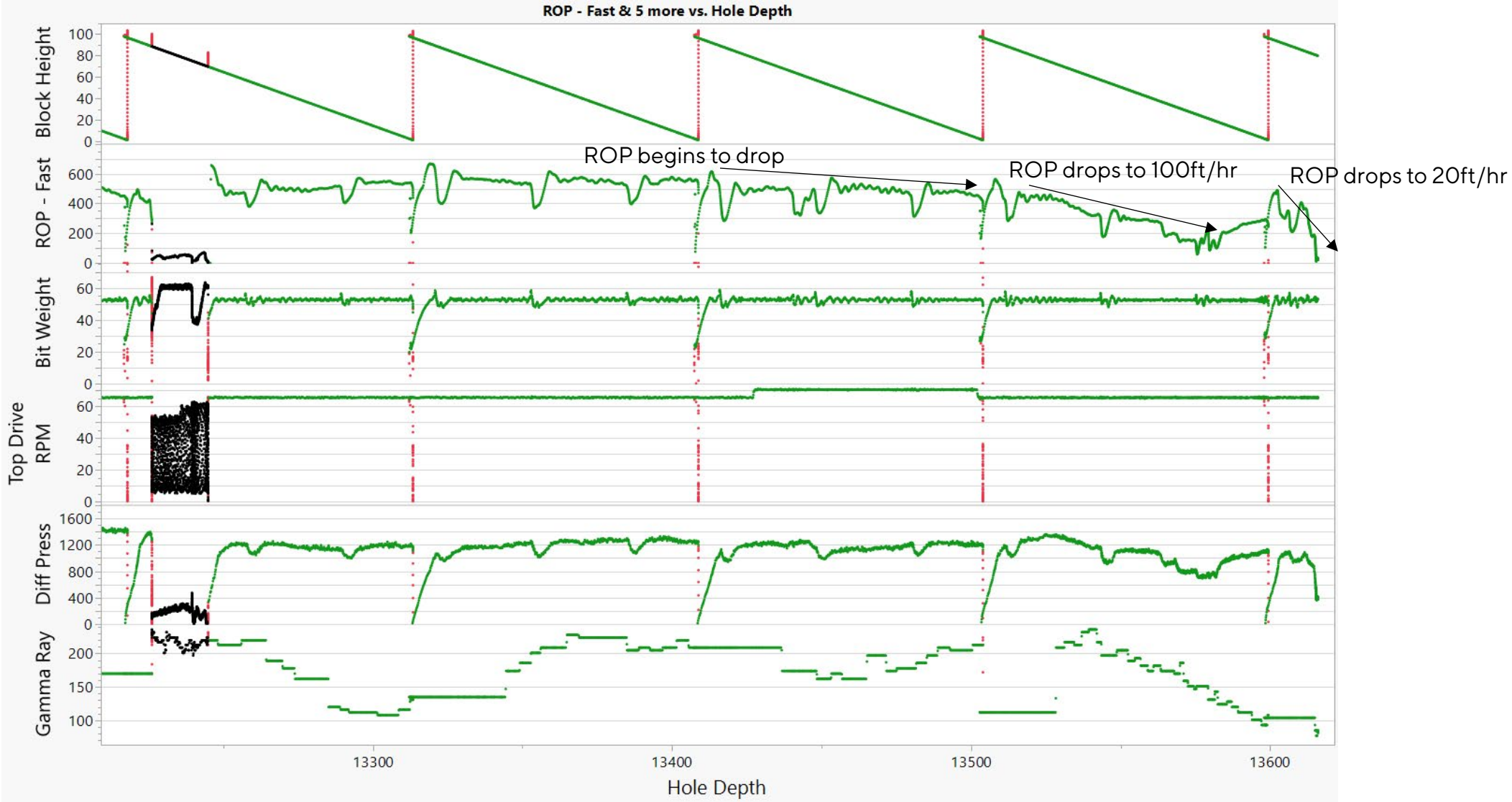
# Last Four Stands Before TOOH

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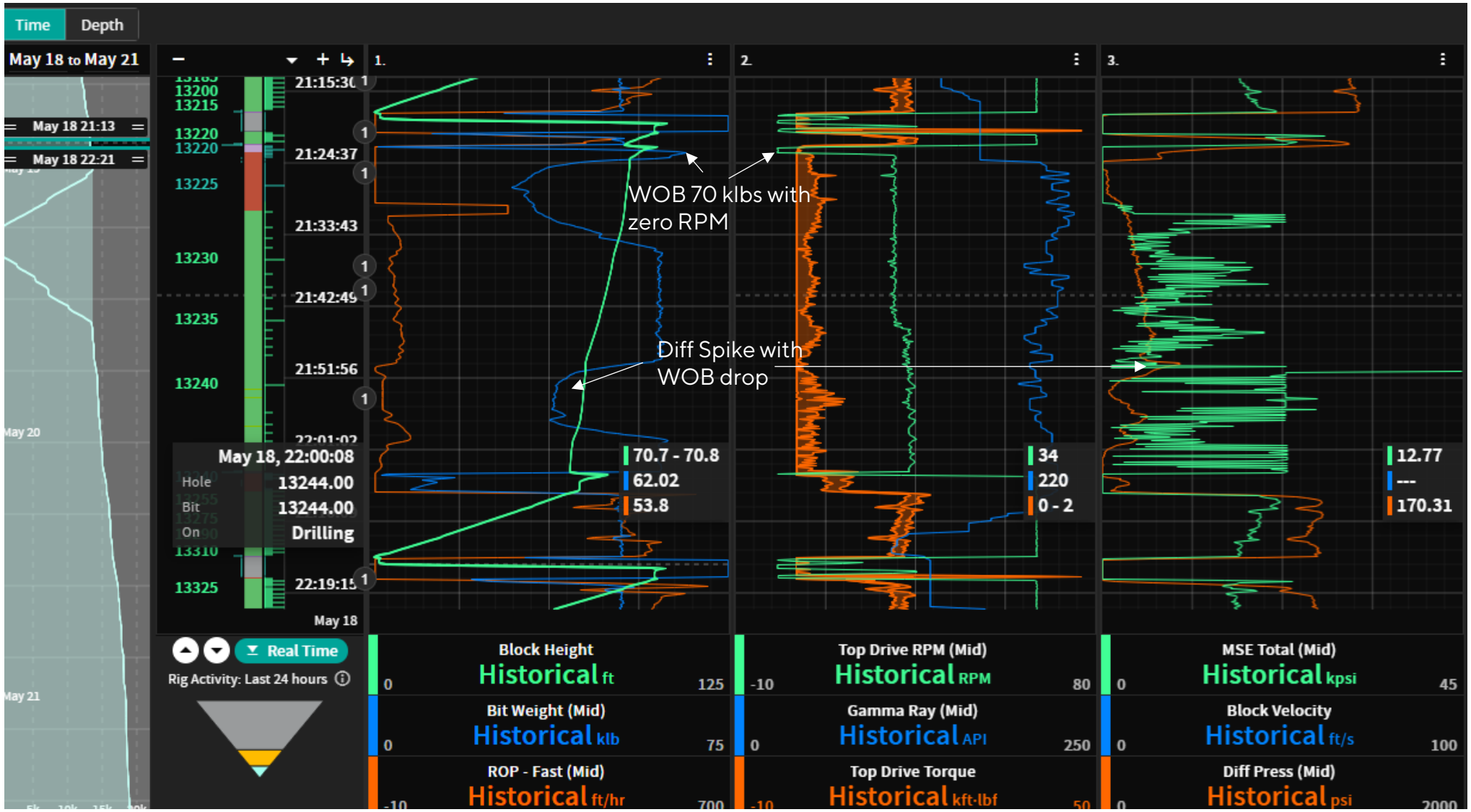
# Last Slide: EDR View



# BHA

| | | Category | Name | OD (in) | ID (in) | Linear Weight (lb/ft) | Length (ft) | Weight (lb) | Cum.Lenth | Grade |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | dp | 5" DP(DS-50) | 5 | 4.28 | 23.75 | 31 | 736.25 | 5242.97 | S135 |
| 2 | | agitator | NOV Shock Sub/Agitator | 6.58 | 2.5 | 99.025 | 26.67 | 2640.997 | 5211.97 | |
| 3 | | dp | 5" DP(DS-50) | 5 | 4.28 | 23.75 | 2572.53 | 61097.588 | 5185.3 | S135 |
| 4 | | agitator | NOV Shock Sub/Agitator | 6.38 | 2.5 | 92.097 | 26.55 | 2445.165 | 2612.77 | |
| 5 | | dp | 5" DP(DS-50) | 5 | 4.28 | 23.75 | 2478.4 | 58862 | 2586.22 | S135 |
| 6 | | dc | NMDC | 6.63 | 3.25 | 89.263 | 30.26 | 2701.105 | 107.82 | |
| 7 | | mwd | MWD | 6.52 | 3.25 | 85.397 | 28.53 | 2436.369 | 77.56 | |
| 8 | | sub | UBHO | 6.52 | 3.25 | 85.397 | 4.85 | 414.174 | 49.03 | |
| 9 | | sub | Float Sub | 6.64 | 2.75 | 97.637 | 4.14 | 404.217 | 44.18 | |
| 10 | | pdm | 6.5" Scout CAM 5/6 9.6 Stg | 6.64 | 3.25 | 89.618 | 39.04 | 3498.684 | 40.04 | |
| 11 | | bit | 7.875" Taurex TSd516 | 7.875 | | | 1 | 100 | 1 | |

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